BRAND · SETTLEMENT FOLD

CONVERGE

THE COMPLETE DOCUMENTATION · v0.1.0

Compile intent into shipped software.

DONE IS PROVEN. — never claimed by the agent.

The dark factory for coding agents: a raw idea enters a **nine-gate production line** and leaves as merged, proven work. Autonomous agents do the building; machines do the checking. One method, one referee CLI, twelve portable skills – **zero runtime dependencies**.

CLAUDE CODE · CODEX · KIMI · GROK BUILD · BASH 3.2+ · MIT

What is Converge? The factory where done is proven.

Converge **compiles intent into software**. Whatever coding agent you run – Claude Code, Codex, Kimi, Grok Build – it works against **signed task specs whose completion is enforced by evals**, not by the agent's word. Every task ships with runnable bash evals authored **before** the work; a task only settles when those evals exit green.

The unit of work is the **Task-Spec**: a self-verifying markdown file with a machine-checked format, an effort budget, an HMAC sign-off seal, and its own definition of done. The `cvg` CLI is the **referee** around it – it frames, dispatches, and gates, but holds zero model credentials and never writes a line of product code.

> **"Agents play; the referee scores."**
>
> PLAIN BASH + STDLIB PYTHON — THE SAME INSTALL SERVES EVERY HARNESS

WHY · 01

The eval decides done

A task cannot settle until its evals exit 0 – completion is a state-machine invariant. The maximal stub attack (all nine specs faked) lands all nine RED.

WHY · 02

The referee is never a player

cvg holds no API keys and calls no models. Engine CLIs authenticate themselves; the referee only frames, dispatches headlessly, and gates.

WHY · 03

Cross-family verification

Tier-2 acceptance sends the diff, the intent, and holdout criteria the builder never saw to a *different vendor's* model, prompted to refute.

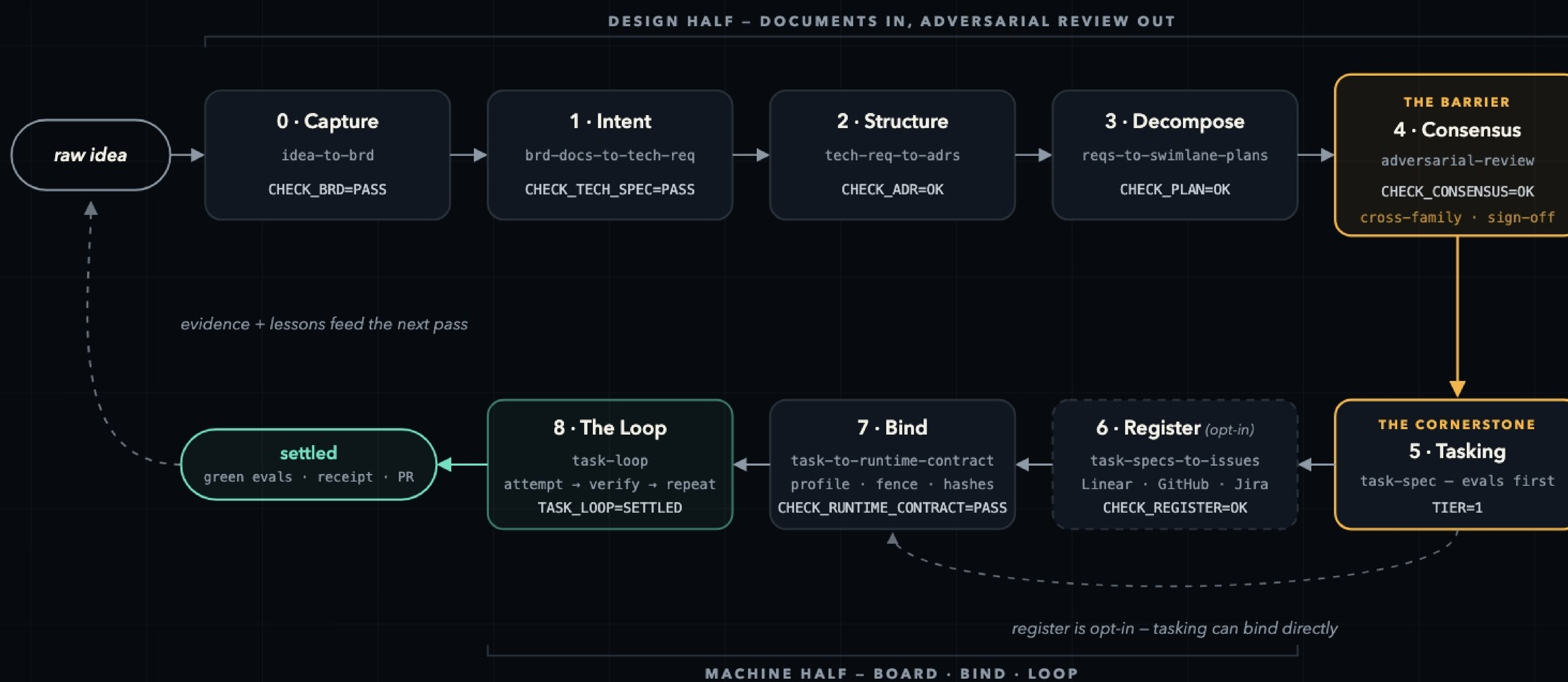
WHY · 04

A loop with brakes

Attempt → verify → repeat under three-axis budgets, a stagnation detector, and eight named terminal states. An exhausted budget is never a success.

9	12	4	0	52	3
SEQUENTIAL GATES	PORTABLE SKILLS	HARNESSES · ONE METHOD	MODEL CREDENTIALS HELD	HERMETIC LOOP CHECKS	LANES · FAST NORMAL FULL

The descent. One job, one output, one gate — per pass.



LANE ROUTING — cvg lane routes work to the ceremony it deserves. It routes, but never waives: nothing irreversible rides FAST, and no lane dispatches an unsigned spec.

FAST	a contained change — task, bind, loop	0	1	2	3	4	**5**	6	**7**	**8**
NORMAL	real features — requirements and structure first	0	**1**	**2**	3	4	**5**	6	**7**	**8**
FULL	new systems — the whole descent, barrier included	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**

5 Tasking — the cornerstone.

SKILL · task-spec | GATE TOKEN · TIER=1 | LANES · FAST + NORMAL + FULL | UPSTREAM · P4 CONSENSUS | DOWNSTREAM · P6 REGISTER / P7 BIND

WHAT YOU DO

Author Task-Specs from the signed plan: **evals first**, then the effort budget, then the HMAC seal. Each spec is atomic, sized on the six-tier scale, and carries its own definition of done.

WHAT THE GATE PROVES

Each spec is **atomic, sized, and safe to delegate** — the sign-off answers `VERDICT: DELEGATE` and stamps the seal. From that byte on, editing an eval breaks the seal.

THE CLI PATH

1. `cvg tasks plan` — read-only preview + rationale
2. `cvg tasks new <slug> <effort>` — scaffold – evals first
3. `cvg tasks validate <spec>` — structure + sizing gate
4. `cvg tasks gate --stamp <spec>` — verdict + HMAC seal
5. `cvg eval <spec>` — run the spec's evals

SEAL=BROKEN If anyone – agent or human – edits an eval after signing, the seal breaks and the gate refuses. Hand-stamping is rejected; the only path to autonomy is through the gate.

cvg/tasks/T-20260730-add-health-endpoint.md — ANATOMY OF A TASK-SPEC

```
1 ---  identity — machine-checked frontmatter
id: T-20260730-add-health-endpoint
effort: XS  # six-tier scale — sized before it is signed
tier: 1     # deterministic evals decide settlement
---

2 ## Evals — authored before the work
```bash
curl -sf localhost:8080/health | grep -q '"ok"'
test "$(curl -so /dev/null -w '%{http_code}' localhost:8080/health)" = 200
```

3 ## Budget — three axes, exhaustion is never success
iterations: 5 · wall-clock: 900s · tokens: capped

4 ## Definition of done — traceability back to the plan
green evals · receipt written · PR opened → TASK_LOOP=SETTLED
```

⑤ SIGNED · HMAC-SHA256 · VERDICT: DELEGATE — seal breaks if one eval byte changes

GATE A · STRUCTURE

`cvg tasks validate <spec>`

Machine-checked format, six-tier sizing, evals present – atomicity before ceremony.

GATE B · SIGN-OFF

`cvg tasks gate --stamp <spec>`

The verdict + the HMAC seal – stamped at the moment a human says "safe to delegate".

One referee. Every verdict ends in one greppable token.

START CONTROL PLANE	`cvg init` `cvg setup` `cvg setup signing` `cvg setup tracker linear` `cvg doctor` `cvg doctor evidence`
DESIGN PASSES 0–4	`cvg capture` `cvg intent` `cvg structure` `cvg decompose` `cvg review --adversary codex` `cvg lane "…"`
AUTHOR & SIGN PASS 5	`cvg tasks plan` `cvg tasks new` `cvg tasks validate` `cvg tasks gate --stamp` `cvg eval` `cvg lint`
BOARD PASS 6 · OPT-IN	`cvg register` `cvg register --check` `cvg ready` `cvg transition`
EXECUTE PASSES 7–8	`cvg bind --task` `cvg loop --issue` `cvg verify --task` `cvg gate --path`
ORIENT FOR YOU + AGENTS	`cvg next` `cvg lesson` `cvg agent-context` `--json` `--dry-run`

WHAT A SESSION ACTUALLY LOOKS LIKE

```
$ cvg tasks gate --stamp cvg/tasks/T-20260730-add-health-endpoint.md
VERDICT: DELEGATE
TIER=1
$ cvg loop --issue T-20260730-add-health-endpoint --agent codex
TASK_LOOP=SETTLED   # exit 0 — only SETTLED · LOCAL_SETTLED · NO_OP exit zero
```

THE TRUST ARCHITECTURE

YOU – intent & judgment
plain language, stage by stage · sign-off at the barrier · objections machines can't settle

↓ steers from chat

HARNESS AGENT
Claude Code · Codex · Kimi · Grok Build – twelve skills installed in each harness's native directory

↓ drives the gates

cvg – THE REFEREE
frames · dispatches headlessly · gates – zero model credentials, never writes product code

↓ signed spec dispatched

THE LOOP – attempt → verify → repeat
engine attempts ⇄ sealed evals verify
bounded: iterations · wall-clock · tokens — stagnation detector armed

↓ only green evals settle

EVIDENCE – the floor the gates stand on TASK_LOOP=SETTLED
receipts · HMAC seals · greppable tokens · tier-2 cross-family verdicts · read-only Cockpit

THE FULL EDITION — WHAT THIS PREVIEW EXPANDS INTO